

IP Address Practice Sheet

IP addressing | IPv4 and IPv6 | Subnetting | Classful and classless addressing

1. Classful and classless IPv4 addressing / subnetting

(a) Under classful IPv4 addressing, state the address class of 172.16.8.20.

(b) State the default subnet mask for that class.

(c) A device has the classless address 192.168.10.70/26. State the subnet mask.

(d) State the network address for 192.168.10.70/26.

(e) State the broadcast address for 192.168.10.70/26.

2. IP address types

Source: 9618/12 M/J/2021 Question 5(d)

Draw one line from each term to its most appropriate description.

Term	Description
Public IP address	It is only visible to devices within the Local Area Network (LAN).
Private IP address	It increments by 1 each time the device connects to the internet.
Dynamic IP address	A new one is reallocated each time a device connects to the internet.
Static IP address	It can only be allocated to a router.
	It is visible to any device on the internet.
	It does not change each time the device connects to the internet.

3. Subnetting benefits, subnet masks, network ID and host ID

Source: 9618/13 O/N/2022 Question 7(c)

Local Area Networks (LANs) can be made up of several subnetworks.

(i) Give two benefits of dividing a network into subnetworks by subnetting the LAN.

1.

2.

(ii) A subnet mask is used when subnetting a LAN. Two devices on the LAN are located in different subnetworks. The IP addresses and corresponding subnet masks are shown:

Device IP address	Subnet mask
10.10.12.1	255.0.0.0
192.168.12.4	255.255.255.0

Identify the following network ID and host ID.

The network ID for the device with the IP address 10.10.12.1

The host ID for the device with the IP address 192.168.12.4

4. IPv4 and IPv6 address structure; benefits of subnetting

Source: 9618/11 M/J/2023 Question 4(d)

The CCTV cameras are connected to a network and transfer their data wirelessly to the central computer.

(i) Each device on the network has an IP address. Complete the description of IP addresses.

An IPv4 address contains _____ groups of digits. Each group is represented in _____ bits and the groups are separated by full stops.

An IPv6 address contains _____ groups of digits. Each group is represented in _____ bits. Multiple groups that only contain zeros can be replaced with a _____.

(ii) The network makes use of subnetting. Describe two benefits of subnetting a network.

1.

2.

5. Network ID and host ID in a subnetwork

Source: 9618/13 M/J/2023 Question 2(e)

One site has split the network into several subnetworks.

An IP address in a subnetwork is divided into two parts.

Identify and describe the two parts of an IP address in a subnetwork.

1.

2.

6. IPv6, subnetting reasons and types of IP address

Source: 9618/12 O/N/2023 Question 7(a), 7(b)(iii), 7(d)

(a) The server has the IP address 192.168.3.2. Explain why this is not an IPv6 address.

(b)(iii) Subnetting can be used to separate a network into logical segments. Describe two other reasons why subnetting is used in a network.

1.

2.

(d) The following incomplete table contains types of IP addresses and their descriptions. Complete the table by writing the missing types of IP addresses and the missing descriptions.

Type of IP address	Description
_____	an IP address that is assigned to a device to allow direct access on the internet
static IP address	_____
_____	an IP address used for internal LAN communication only
dynamic IP address	_____

7. Public IP addresses and IPv4/IPv6 differences

Source: 9618/11 M/J/2024 Question 8(b)

The LAN will connect to the internet through a router. The router has a public IPv6 address.

(i) State why the router has a public IP address.

(ii) One difference between an IPv4 and IPv6 address is that the numbers in an IPv4 address are separated by full stops and in an IPv6 address they are separated by colons. Identify two other differences between an IPv4 and IPv6 address.

1.

2.

8. Static private IP addresses

Source: 9618/13 M/J/2024 Question 5(d)

The devices in the office have static private IP addresses.

State what is meant by a static private IP address.

9. Purpose of public/private IP addresses and subnetting

Source: 9618/13 O/N/2024 Question 9(d)

The LAN has a router. The router has a public IP address and a private IP address.

(i) State the purpose of a public IP address and a private IP address.

Public IP address:

Private IP address:

(ii) The LAN uses subnetting. Describe the purpose of subnetting in a network.

10. Complete the IPv4/IPv6 and IP-address-type description

Source: 9618/12 M/J/2025 Question 6(c)

Devices that connect to the WAN have an IP address. Complete the following description of IP addresses by writing the missing words.

IPv4 is displayed as four groups of 8-bit numbers separated by _____.

Each IPv4 address is 32 bits.

IPv6 is made of eight groups of 4 _____ numbers separated by colons.

Multiple consecutive groups of _____ can be replaced with a double colon.

Each IPv6 address is _____ bits.

A _____ IP address can change each time the computer connects to a network.

A _____ IP address can only be accessed by other devices in the same LAN and is assigned by the router within the LAN.

11. Static/public IP addresses and IP-address validity

Source: 9618/11 O/N/2025 Question 7(a-b)

Web browsers use Internet Protocol (IP) addresses.

(a) Complete the table by describing the following types of IP address.

Type of IP address	Description
Static	
Public	

(b) Consider the following IP address:

256.0.0.A

Circle whether this IP address is IPv4, IPv6 or an invalid IP address.

- ☐ IPv4 ☐ IPv6 ☐ Invalid

Justify your choice.

12. Static private IP addresses and benefits of subnetting

Source: 9618/11 M/J/2026 Question 7(c)

Each of the school's computers has a static private IP address.

(i) State what is meant by a static private IP address.

(ii) The school's network uses subnetting. Give three benefits of using subnetting in a network.

1.

2.

3.

13. Router tasks involving IP addresses

Source: 9618/11 M/J/2021 Question 4(c)(i) and 9618/13 M/J/2021 Question 4(c)(i) - identical item

Tick one box in each row to identify whether the task is performed by the router or not.

Task	Performed by router	Not performed by router
Receives packets from devices		
Finds the IP address of a Uniform Resource Locator (URL)		
Directs each packet to all devices attached to it		
Stores the IP and/or MAC address of all devices attached to it		